

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

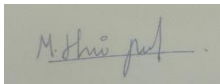
Question Count: 5

Code of Conduct

I Hariprasath(192421246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1.Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Type-1 Hypervisor	Type-2 Hypervisor
Runs directly on the hardware.	Runs on top of a host operating system.
Gives better speed and performance.	Performance is slower because it depends on the host OS.
More secure since there is no extra operating system layer.	Less secure because if the host OS has a problem, the VMs may also be affected.
Used in cloud data centers and enterprises.	Mostly used for learning, testing, and personal use.
Example: VMware ESXi, Hyper-V, KVM.	Example: VirtualBox, VMware Workstation.

Performance

Type-1 hypervisors provide better performance because they communicate directly with the hardware. This reduces delays and makes them suitable for running many virtual machines. Type-2 hypervisors are slower since all requests go through the host operating system.

Security

Type-1 hypervisors are more secure because there is no host operating system that attackers can target. In Type-2 hypervisors, if the host operating system is infected or crashes, the virtual machines can also be affected.

Manageability

Type-1 hypervisors are easier to manage in large cloud environments. They support features like live migration, load balancing, snapshots, and high availability. Type-2 hypervisors are simple to install and use but do not provide many advanced management features.

Justification

For a cloud datacenter that hosts **mission-critical applications**, **Type-1 hypervisors are the better choice**. They offer better performance, stronger security, and reliable management features. They also ensure high availability and can handle heavy workloads efficiently. Type-2 hypervisors are mainly suitable for testing, development, and personal use.

Conclusion

Therefore, **Type-1 hypervisors are the best option for cloud computing** because they are faster, more secure, and more reliable for running important business applications.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Introduction

Virtualization is an important technology in cloud computing. It allows multiple virtual machines (VMs) to run on a single physical server. Each VM acts like an independent computer with its own operating system and applications. This helps cloud providers host different workloads on the same hardware while keeping them isolated from one another.

How Virtualization Supports Isolation

1. Separate Virtual Machines

- Each application, such as finance, healthcare, and student portals, runs in its own VM.
- A problem in one VM does not affect the others.

2. Resource Isolation

- The hypervisor allocates CPU, memory, storage, and network resources separately for each VM.
- This prevents one workload from using all the resources.

3. Security Isolation

- Each VM has its own operating system and security settings.
- Users in one VM cannot directly access data stored in another VM.

4. Fault Isolation

- If one VM crashes or is infected with malware, the other VMs continue to run normally.

Two Major Attack Surfaces

1. Hypervisor Attack

- The hypervisor controls all virtual machines.
- If an attacker exploits a vulnerability in the hypervisor, they may gain access to multiple VMs and their data.

2. VM Escape Attack

- In a VM escape attack, malware inside one VM breaks out of its virtual environment and tries to access the host system or other VMs.
- This can lead to data theft or unauthorized access.

Conclusion

Virtualization provides strong isolation by running each workload in separate virtual machines while sharing the same physical hardware. It improves security, resource utilization, and reliability. However, cloud providers must protect the **hypervisor** and prevent **VM escape attacks** to ensure the safety of sensitive applications like finance, healthcare, and student portals.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, Memory 88%, Storage I/O Latency 35 ms, Average VM Boot Time 170 s. Explain the likely bottlenecks.

Introduction

The given VM host utilization shows that the server is under heavy load. High CPU and memory usage, along with high storage latency and slow VM boot time, indicate performance bottlenecks. These issues can reduce the efficiency of the cloud environment and affect user experience.

Likely Bottlenecks

1. High CPU Utilization (92%)

- CPU usage is very high, which means the processor is handling too many tasks.
- Virtual machines may become slow and take longer to process requests.

2. High Memory Utilization (88%)

- Most of the RAM is already in use.
- The system may start swapping memory to disk, which decreases performance.

3. High Storage I/O Latency (35 ms)

- Storage latency is much higher than the normal range (usually below 10 ms).
- Reading and writing data becomes slow, affecting application performance.

4. Slow VM Boot Time (170 seconds)

- A VM takes nearly 3 minutes to start, which is longer than expected.
- This is mainly caused by high CPU load, insufficient memory, and slow storage access.

Corrective Actions

1. Reduce CPU Load

- Migrate some VMs to another host using live migration.
- Add more CPU cores or upgrade the processor if required.

2. Increase Memory

- Allocate more RAM to the host.
- Remove unnecessary applications or optimize memory usage.

3. Improve Storage Performance

- Replace HDDs with SSDs or NVMe storage.

- Use faster storage arrays and optimize disk I/O.

4. **Balance the Workload**

- Distribute virtual machines across multiple hosts.
- Enable load balancing to avoid overloading a single server.

5. **Monitor Resource Usage**

- Continuously monitor CPU, memory, and storage using cloud monitoring tools.
- Take preventive actions before resources become overloaded.

Conclusion

The main bottlenecks are **high CPU usage (92%)**, **high memory utilization (88%)**, **high storage I/O latency (35 ms)**, and **slow VM boot time (170 s)**. These problems can be solved by increasing hardware resources, improving storage performance, balancing workloads, and monitoring the cloud infrastructure regularly. This will improve overall system performance and reduce VM startup time.

4. Analyze how Software Defined Datacenter (SDDC) architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Introduction

A **Software Defined Datacenter (SDDC)** is a modern data center where compute, storage, and networking are managed through software instead of physical hardware. This makes the data center more flexible, automated, and easier to manage. Compared to a traditional data center, an SDDC can quickly respond to changing business needs.

How SDDC Improves Agility

1. Faster Resource Provisioning

- In a traditional data center, configuring servers and networks takes time and manual effort.
- In an SDDC, resources can be created and managed automatically through software within minutes.

2. Automation

- Most management tasks, such as deploying virtual machines, configuring storage, and setting up networks, are automated.
- This reduces manual work and human errors.

3. Better Resource Utilization

- Resources are allocated based on demand.
- Unused resources can be assigned to other applications, improving efficiency.

Virtualization Considerations

1. Compute Virtualization

- Physical servers are divided into multiple virtual machines (VMs).
- Applications can run independently on different VMs.
- If one server becomes overloaded, VMs can be moved to another server using live migration without interrupting services.

2. Storage Virtualization

- Multiple storage devices are combined into a single virtual storage pool.
- Storage can be allocated or expanded whenever needed without changing physical hardware.
- This improves storage management and availability.

3. Network Virtualization

- Virtual networks are created using software instead of physical switches and routers.
- Network configurations can be changed quickly based on application requirements.
- It also improves security by isolating different workloads.

Comparison with Traditional Datacenter

Traditional Datacenter	Software Defined Datacenter (SDDC)
Hardware-based management	Software-based management
Manual configuration	Automated configuration
Slow resource provisioning	Fast resource provisioning
Limited flexibility	Highly flexible and scalable
Lower resource utilization	Better resource utilization

Conclusion

A Software Defined Datacenter is more agile than a traditional data center because it uses **compute, storage, and network virtualization** to automate resource management. It provides faster deployment, better scalability, improved resource utilization, and easier management, making it the preferred choice for modern cloud computing environments.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Introduction

A **multi-cloud federation** allows organizations to use services from different cloud providers. For this to work properly, users should be able to access all cloud services using the same identity. If different cloud providers use different identity systems, authentication may fail, causing users to lose access to resources.

Analysis of the Issue

The project fails because the cloud providers cannot recognize or verify the same user identity. This may happen due to:

1. Different user IDs and authentication methods.
2. Different identity management systems used by each provider.
3. Lack of trust between cloud providers.
4. Incompatible security policies and access permissions.

As a result, users may be denied access, required to log in multiple times, or experience security risks.

Secure Identity and Privacy Design

1. Identity Federation

- Use a trusted **Identity Provider (IdP)** that authenticates users once.
- All cloud providers trust the IdP and accept its authentication.

2. Single Sign-On (SSO)

- Users log in only once and can access services across different cloud providers without repeated authentication.
- This improves user experience and security.

3. Standard Authentication Protocols

- Use standard protocols such as **SAML**, **OAuth 2.0**, or **OpenID Connect** for secure identity sharing between providers.
- These standards ensure compatibility among different cloud platforms.

4. Role-Based Access Control (RBAC)

- Users are assigned roles such as Admin, Employee, or Student.
- Each role receives only the permissions required to perform its tasks.

5. Privacy Protection

- Share only the minimum user information needed for authentication.
- Encrypt identity data during transmission and storage.
- Enable Multi-Factor Authentication (MFA) for additional security.

Benefits of Federation

- Single login for multiple cloud services.
- Better security and privacy.
- Reduced password management.
- Easier user administration across cloud providers.
- Improved trust between different cloud environments.

Conclusion

Identity mismatches are a major challenge in multi-cloud environments. By using **Identity Federation, Single Sign-On (SSO)**, standard authentication protocols like **SAML, OAuth, or OpenID Connect**, **Role-Based Access Control (RBAC)**, and strong privacy measures, cloud providers can ensure secure and seamless access across multiple cloud platforms. This improves both security and user experience.